

- Q-1 Reduce using K-Map: $f = \prod_M(2, 8, 9, 10, 11, 12, 14)$ [Ans. $f = (A'+B) (A'+D) (B+C'+D)$]
- Q-2 Reduce using K-Map: $f = \prod_M(0, 1, 2, 6, 8, 10, 11, 12)$ [Ans. $f = (A'+C+D) (A'+B+C') (A+B+C) (A+C'+D)$]
- Q-3 Reduce using K-Map: $f = \sum_m(1, 4, 7, 10, 13) + d(5, 14, 15)$ [Ans. $f = BD + A'BC' + A'C'D + ACD'$]
- Q-4 Reduce using K-Map: $f = \sum_m(4, 5, 7, 12, 14, 15) + d(3, 8, 10)$ [Ans. $f = AD' + A'BC' + BCD$]
- Q-5 Reduce using K-Map: $f = \sum_m(6, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$ [Ans. $f = A + BC$]
- Q-6 Reduce using K-Map: $f = \prod_M(1, 4, 5, 11, 12, 14) \cdot d(6, 7, 15)$ [Ans. $f = (B'+D) (A+C+D') (A'+C'+D')$]
- Q-7 Reduce using K-Map: $f = \prod_M(3, 6, 8, 11, 13, 14) \cdot d(1, 5, 7, 10)$
[Ans. $f = (A'+B+D) (B'+C+D') (B'+C'+D) (B+C'+D')$]
- Q-8 Reduce using K-Map: $f = \sum_m(0, 1, 4, 5, 6, 7, 9, 11, 15) + d(10, 14)$ [Ans. $f = A'C' + BC + AB'D$]
- Q-9 Reduce using K-Map: $f = \sum_m(9, 10, 12) + d(3, 5, 6, 7, 11, 13, 14, 15)$ [Ans. $f = AB + AD + AC$]
- Q-10 Reduce using K-Map: $f = \sum_m(0, 2, 3, 4, 7, 9, 15) + d(6, 8, 11)$ [Ans. $f = CD + A'D' + AB'C'$]
- Q-11 Reduce using K-Map: $f = \sum_m(1, 5, 6, 12, 13, 14) + d(2, 4)$ [Ans. $f = BD' + BC' + A'C'D$]